Omar Alama

PhD Student in Robotics & Computer Vision · Carnegie Mellon University

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Research interests: robots that understand and act in the open world via agentic systems and vision-language-action models for semantic navigation, open-vocabulary 3D scene understanding and mapping, and efficient multimodal foundation models for spatial reasoning.

Education

Carnegie Mellon University — Pittsburgh, PA
Ph.D. in Electrical and Computer Engineering · AirLab, Robotics Institute — Aug 2024 – Present
Advisor: Prof. Sebastian Scherer

M.S. in Electrical and Computer Engineering · GPA 4.0/4.0 — Aug 2022 – May 2024
Advisor: Prof. Marios Savvides

King Abdulaziz University (KAU) — Jeddah, Saudi Arabia
B.S. in Computer Engineering · GPA 5.0/5.0 · Valedictorian — Aug 2015 – May 2020

Experience

Amazon Lab126 — Sunnyvale, CA
Applied Scientist Intern II — May 2026 – Present

- Researching 3D spatial scene understanding with multimodal large language models (MLLMs).

Carnegie Mellon University – AirLab, Robotics Institute — Pittsburgh, PA
Graduate Research Assistant — Jan 2024 – Present

- Developing real-time, open-vocabulary semantic scene representations for UAVs and UGVs operating in unstructured environments [1,3,4].
- Designing parameter- and compute-efficient zero-shot open-vocabulary segmentation built on agglomerative vision foundation models [1].
- Developing and field-testing mapping and navigation algorithms for multi-drone search-and-rescue systems [2].
- Exploring efficient scene–LLM integration for stronger spatial reasoning and understanding.

Carnegie Mellon University – AI for Vision and Biometrics Center — Pittsburgh, PA
Graduate Research Assistant — Jan 2023 – May 2024

- Led the design of a proprietary multi-camera auto-checkout system, solving multi-view product tracking and re-identification.
- Built a semi-automated 3D scanning pipeline (DSLR triggering, segmentation, 3D reconstruction) and a Blender-based procedural generator that synthesized a 100K+ image segmentation and detection dataset, used to train a robust produce detector.
- Designed and built a 48×RTX 4090 GPU compute cluster, saving the lab hundreds of thousands of dollars over commercial servers.

King Abdullah University of Science and Technology (KAUST) — Thuwal, Saudi Arabia
Research Software Engineer (advisor: Marco Canini) — Sep 2020 – Oct 2021

- Consolidated and refactored 4+ separately developed codebases into the public release of SwitchML (NSDI'21), delivering further speedups and bug fixes.
- Wrote C++ CPU and CUDA GPU kernels integrated with PyTorch distributed training to evaluate the training speedup of in-network floating-point operations on programmable switches [5].

Research Intern — May 2019 – Aug 2019

- Built a SimPy simulator of distributed deep learning communication to evaluate all-reduce scheduling.

Saudi Company for Artificial Intelligence (SCAI) — Riyadh, Saudi Arabia
Software Engineer — Oct 2021 – Dec 2021

- Led a team of 3 integrating breast cancer detection models into hospital radiology workflows via DICOM.

King Abdulaziz University (KAU) — Jeddah, Saudi Arabia
Lecturer — Dec 2021 – Jul 2022

- Taught structured programming in MATLAB to 90 students and ran Data Structures & Algorithms labs in Java.

Selected Publications

* denotes equal contribution

[1] RADSeg: Unleashing Parameter and Compute Efficient Zero-Shot Open-Vocabulary Segmentation Using Agglomerative Models
O. Alama*, D. Jariwala*, A. Bhattacharya*, S. Kim, W. Wang, S. Scherer
CVPR 2026 (Findings); also at the CVPR 2026 OpenSUN3D Workshop [Project] [arXiv] [Code] [Demo]

[2] RAVEN: Resilient Aerial Navigation via Open-Set Semantic Memory and Behavior Adaptation
S. Kim, **O. Alama**, D. Kurdydyk, J. Keller, N. Keetha, W. Wang, Y. Bisk, S. Scherer
ICRA 2026 · **Oral (8.3% of accepted papers)** · **Best Paper Finalist**, IROS'25 Active Perception Workshop
[Project] [arXiv] [Code] [Video]

[3] RayFronts: Open-Set Semantic Ray Frontiers for Online Scene Understanding and Exploration
O. Alama, A. Bhattacharya, H. He, S. Kim, Y. Qiu, W. Wang, C. Ho, N. Keetha, S. Scherer
IROS 2025 [Project] [arXiv] [Code]

[4] Map It Anywhere: Empowering BEV Map Prediction using Large-scale Public Datasets
C. Ho*, J. Zou*, **O. Alama***, S. Kumar, B. Chiang, T. Gupta, C. Wang, N. Keetha, K. Sycara, S. Scherer
NeurIPS 2024, Datasets and Benchmarks Track [Project] [arXiv] [Code]

[5] Unlocking the Power of Inline Floating-Point Operations on Programmable Switches
Y. Yuan, **O. Alama**, J. Fei, J. Nelson, D. R. K. Ports, A. Sapio, M. Canini, N. S. Kim
NSDI 2022 [Paper]

[6] Aug3D: Augmenting Large Scale Outdoor Datasets for Generalizable Novel View Synthesis
A. Rauniyar*, **O. Alama***, S. Yong, K. Sycara, S. Scherer
IROS 2024 Workshop [Project] [arXiv]

Honors & Awards

- **Oral Presentation**, ICRA 2026 (8.3% of accepted papers) – RAVEN — 2026
- **Best Paper Finalist** and Spotlight Presentation, IROS 2025 Active Perception Workshop – RAVEN — 2025
- **Valedictorian**, ranked 1st in engineering cohort at KAU; awarded a scholarship for graduate studies — 2020
- **Commercialization agreement** from KAU for capstone project Hadir (face-recognition attendance) — 2020

Selected Coursework

Carnegie Mellon University: Computer Vision, Learning for 3D Vision, Visual Learning & Recognition, Robot Localization and Mapping (SLAM), Geometry-Based Methods in Vision, Planning and Decision-Making in Robotics, Autonomous Driving, Pattern Recognition Theory, Optimization, Applied Stochastic Processes, Embedded Deep Learning, Systems and Toolchains for AI

Skills

Research	Robotics, semantic navigation and exploration, agentic systems, VLAs, MLLMs, vision foundation models, open-vocabulary segmentation and 3D mapping, 3D reconstruction, synthetic data
Engineering	Python, PyTorch, C++, CUDA, distributed training, Docker, Git, Blender
Languages	English (fluent), Arabic (native)